$$\frac{CD}{BD}=\frac{AF}{AE}=\frac{r_2\cos\varphi_2}{r_1\cos\varphi_1},\qquad (6a)$$

Substituting (3a)~(6a) into gives

$$\left(\frac{r_1}{r_2}\right)^{\alpha-1}=\frac{r_1\sin\varphi_1}{r_2\sin\varphi_2}=\frac{r_2\cos\varphi_2}{r_1\cos\varphi_1}。\qquad (7a)$$

From (7a),

$$\sin^{\alpha}\varphi_1\cos^{\alpha-2}\varphi_1=\sin^{\alpha}\varphi_2\cos^{\alpha-2}\varphi_2。\qquad (8a)$$

The remaining argument is the same as in "Method 1".

Scoring: see "Method 1".

V. (15 points)

Ambient pressure is p_a. Initially the air has pressure p_a, temperature T_a, volume V_a. Contact with the iceberg brings equilibrium at temperature T_1 and volume V_0. This is an isobaric process, so

$$\frac{V_0}{T_1}=\frac{V_a}{T_a}\qquad (1)$$

The change in the air's internal energy during this process is

$$\Delta U=2.5p_a\left(V_0-V_a\right),\qquad (2)$$

The work done by the atmosphere on the air is

$$W=-p_a\left(V_0-v_a\right)\qquad (3)$$

Let Q be the heat from the iceberg to the air. The first law of thermodynamics gives

$$Q=\Delta U-W。\qquad (4)$$

The four preceding equations give

$$Q=3.5p_aV_a\left(\frac{T_1-T_a}{T_a}\right)\qquad (5)$$

In (5), Q is negative: heat actually flows from the air to the iceberg.

After cooling to the ice temperature, the air is heated at fixed volume V_0, from T_1 to ambient temperature T_a, absorbing heat from its surroundings. If pressure is p_1,

$$\frac{p_1}{T_a}=\frac{p_a}{T_1}\qquad (6)$$

Let u be the nozzle volume. The first jet lowers vessel pressure from p_1 to p_2. The given conditions imply

$$p_1\left(V_0-u\right)=p_2V_0,\qquad (7)$$